RetinAI ROP Staging: A Results Compendium of Every Model Trained (2026)

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***Abstract*—This document reports, without discussion, every model the RetinAI project trained for staging retinopathy of prematurity (ROP) in 2026, the protocol each was scored under, and the file each number comes from. The current generation (9 arms, A–I) shares one pre-registered protocol: patient-grouped five-fold cross-validation (CV) on 6,700 photographs of 1,528 infants from 3 hospitals, eye-level scoring at a fourth hospital unseen in training, and leave-one-source-out folds. The selected model, B (DINOv2 ViT-S/14), scores CV QWK5 0.825 [0.737, 0.911] and, at the unseen hospital, eye-level QWK5 0.756 [0.679, 0.821]. Earlier generations are tabulated separately under their own protocols.**

***Index Terms*—retinopathy of prematurity, ICROP-3, staging, benchmark, results, pre-registration**

I. How to Read

Settings. (i) *CV per photograph*: five patient-grouped folds, out-of-fold, seeds 42 and 1337 pooled, 14 duplicates excluded. (ii) *Unseen hospital per eye*: Multi-View development half (663 photographs, 230 eyes); eye = most severe predicted stage; descriptive; locked half unopened for A–I. (iii) *LOSO*: train on two sources, score the third. **Metrics.** QWK5: quadratic-weighted kappa, five ordinal levels, AP-ROP folded onto stage 4/5 [1]; mF1: mean per-class F1, present classes; Acc.: exact six-class matches; CIs: 95% percentile bootstrap, 2,000 infant resamples. **Pre-registration.** `PROTOCOL_V2.md` (`ac33f29`, 2026-10-20; 5 amendments to 2026-10-31); selection on CV only. **Caveats** (`MODEL_HISTORY.md`): QWK definitions, eye rules and test halves changed across generations; the June binary numbers carry a device shortcut. Section III is never pooled with A–I.

II. Current Generation (Protocol v2, Arms A–I)

Tables I–VIII and Figs. 1–9. Arms: A RETFound-Green [2], B DINOv2 [3], C EfficientNetV2-S [4] from [5], D–I variants of B (Table VII).

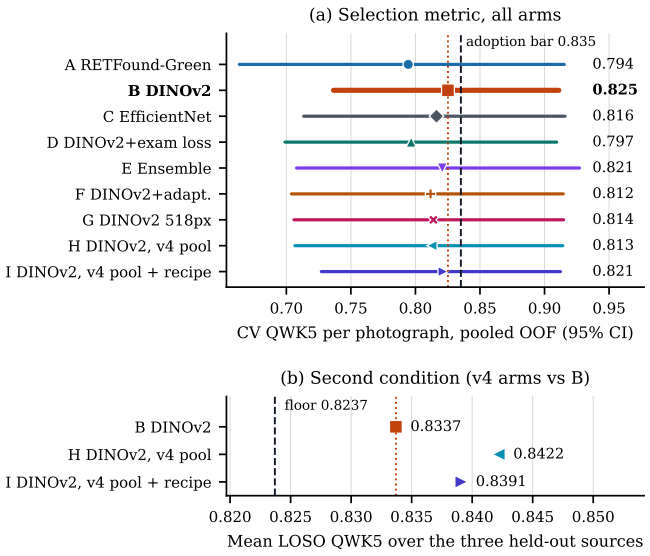


Fig. 1. Selection. (a) CV QWK5 per photograph, pooled out-of-fold, 95% CI. Dotted: B, chosen by the v3 rule. Dashed: the Amendment 5c adoption bar (B + 0.010) a v4 arm (H, I) had to reach. (b) Its second condition: mean LOSO QWK5 over three held-out sources at least B − 0.010 (dashed). No v4 arm met both; B stays.

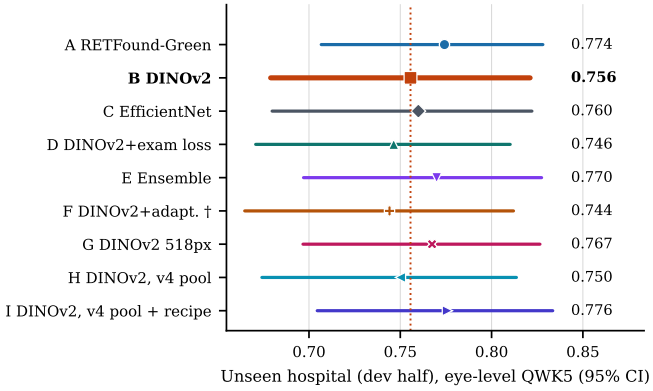


Fig. 2. Unseen hospital, development half: eye-level QWK5, 95% CI, 10 fold-seed models per arm. Descriptive; selected nothing. †Transductive.

TABLE I

HEADLINE RESULTS, ALL ARMS, BOTH SEEDS POOLED. CV: POOLED OUT-OF-FOLD, PER PHOTOGRAPH. DEV: UNSEEN HOSPITAL, DEVELOPMENT HALF, PER EYE. QWK5 WITH 95% INFANT-CLUSTERED CI; ACC. SIX-CLASS ACCURACY; BAL. ACC. MEAN PER-CLASS RECALL; MF1 MACRO-F1 OVER PRESENT CLASSES; AUC AND AP: ANY ROP VS NORMAL (MEAN OVER EACH ARM'S MODELS). BOLD LETTER: SELECTED ARM. †TRANSDUCTIVE: SAW THE DEV PHOTOGRAPHS UNLABELLED IN TRAINING.

		CV, per photograph (6,686 photos)					Unseen hospital, dev half, per eye (230 eyes)				
Arm	Model	QWK5 [95% CI]	Acc.	Bal. acc.	mF1	AUC	QWK5 [95% CI]	Acc.	mF1	AUC	AP
A	RETFound-Green ViT-S	0.794 [0.664, 0.915]	0.759	0.656	0.643	0.977	0.774 [0.707, 0.828]	0.632	0.563	0.935	0.984
B	DINOv2 ViT-S	0.825 [0.737, 0.911]	0.766	0.674	0.653	0.979	0.756 [0.679, 0.821]	0.630	0.566	0.937	0.984
C	EfficientNetV2-S (baseline)	0.816 [0.713, 0.916]	0.788	0.672	0.656	0.967	0.760 [0.680, 0.822]	0.624	0.573	0.943	0.986
D	DINOv2 + exam-level loss	0.797 [0.699, 0.909]	0.803	0.671	0.661	0.942	0.746 [0.671, 0.810]	0.609	0.542	0.935	0.984
E	Ensemble A+B+C	0.821 [0.708, 0.927]	0.787	0.680	0.664	0.981	0.770 [0.697, 0.827]	0.639	0.568	0.944	0.986
F†	DINOv2 + adaptation	0.812 [0.704, 0.915]	0.761	0.663	0.644	0.980	0.744 [0.665, 0.812]	0.624	0.552	0.940	0.985
G	DINOv2 at 518 px	0.814 [0.706, 0.915]	0.764	0.667	0.647	0.984	0.767 [0.697, 0.827]	0.627	0.551	0.939	0.985
H	DINOv2 on the v4 pool	0.813 [0.707, 0.914]	0.765	0.660	0.646	0.982	0.750 [0.674, 0.814]	0.620	0.556	0.941	0.985
I	DINOv2, v4 pool + v4 recipe	0.821 [0.727, 0.912]	0.750	0.645	0.618	0.982	0.776 [0.705, 0.834]	0.633	0.561	0.944	0.986

TABLE II

RECALL / PRECISION PER CLASS, BOTH SEEDS POOLED. UPPER BLOCK: CV PER PHOTOGRAPH; LOWER BLOCK: UNSEEN HOSPITAL (DEV HALF) PER EYE. CLASS COUNTS IN EACH BLOCK'S HEADER. CLASSES [1]: S STAGE, AP AGGRESSIVE ROP.

Arm	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP
CV, per photograph; n photos: Normal 3,837, S1 574, S2 952, S3 617, S4/5 89, AP 617						
A	0.912 / 0.947	0.569 / 0.293	0.584 / 0.541	0.725 / 0.848	0.876 / 0.741	0.268 / 0.760
B	0.909 / 0.959	0.583 / 0.302	0.561 / 0.583	0.722 / 0.788	0.882 / 0.668	0.389 / 0.705
C	0.951 / 0.932	0.497 / 0.369	0.604 / 0.565	0.724 / 0.810	0.876 / 0.596	0.382 / 0.801
D	0.993 / 0.893	0.475 / 0.582	0.488 / 0.646	0.712 / 0.772	0.865 / 0.553	0.490 / 0.581
E	0.943 / 0.949	0.551 / 0.346	0.620 / 0.570	0.729 / 0.847	0.910 / 0.684	0.328 / 0.748
F†	0.911 / 0.959	0.570 / 0.291	0.562 / 0.566	0.722 / 0.767	0.888 / 0.706	0.327 / 0.746
G	0.914 / 0.962	0.607 / 0.308	0.567 / 0.552	0.720 / 0.802	0.893 / 0.720	0.302 / 0.705
H	0.924 / 0.952	0.570 / 0.326	0.557 / 0.520	0.725 / 0.807	0.876 / 0.740	0.306 / 0.682
I	0.909 / 0.960	0.490 / 0.298	0.568 / 0.497	0.741 / 0.725	0.893 / 0.677	0.269 / 0.617
Unseen hospital (dev half), per eye; n eyes: Normal 50, S1 68, S2 46, S3 59, S4/5 5, AP 2						
A	0.930 / 0.620	0.521 / 0.595	0.544 / 0.621	0.585 / 0.888	0.540 / 0.725	0.650 / 0.106
B	0.948 / 0.623	0.510 / 0.607	0.546 / 0.666	0.566 / 0.867	0.580 / 0.760	0.700 / 0.088
C	0.972 / 0.633	0.404 / 0.621	0.591 / 0.544	0.598 / 0.883	0.660 / 0.704	0.850 / 0.135
D	0.956 / 0.606	0.482 / 0.591	0.517 / 0.632	0.537 / 0.867	0.540 / 0.768	0.600 / 0.067
E	0.974 / 0.627	0.491 / 0.623	0.570 / 0.625	0.586 / 0.887	0.560 / 0.718	0.650 / 0.096
F†	0.974 / 0.626	0.484 / 0.633	0.546 / 0.654	0.559 / 0.867	0.540 / 0.717	0.600 / 0.064
G	0.962 / 0.624	0.507 / 0.597	0.522 / 0.639	0.571 / 0.899	0.540 / 0.700	0.550 / 0.070
H	0.950 / 0.625	0.497 / 0.600	0.539 / 0.622	0.551 / 0.864	0.580 / 0.802	0.550 / 0.071
I	0.936 / 0.649	0.506 / 0.648	0.596 / 0.628	0.559 / 0.821	0.560 / 0.783	0.550 / 0.067

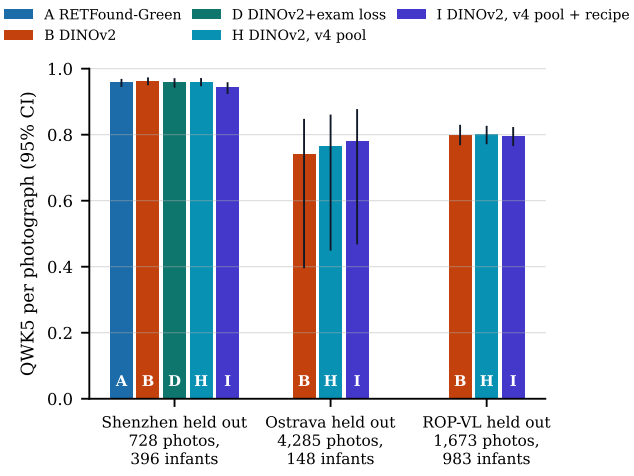


Fig. 3. LOSO QWK5 per photograph with 95% CI (Table IV); bars labelled by arm. Fold compositions in Table XVI.

TABLE III

CV QWK5 PER PHOTOGRAPH PER FOLD (F0–F4) AND SEED, EACH FOLD SCORED BY ITS OWN MODEL; MEAN AND SD OVER THE FIVE FOLDS.

Arm	Seed	F0	F1	F2	F3	F4	Mean	SD
A	42	0.623	0.868	0.872	0.907	0.918	0.838	0.122
	1337	0.595	0.868	0.882	0.911	0.916	0.834	0.135
B	42	0.743	0.870	0.853	0.904	0.898	0.854	0.065
	1337	0.666	0.879	0.857	0.923	0.909	0.847	0.104
C	42	0.680	0.865	0.889	0.899	0.926	0.852	0.099
	1337	0.667	0.863	0.853	0.934	0.917	0.847	0.106
D	42	0.691	0.851	0.810	0.897	0.882	0.826	0.083
	1337	0.650	0.858	0.816	0.920	0.883	0.826	0.105
E	42	0.682	0.874	0.905	0.918	0.929	0.862	0.102
	1337	0.634	0.881	0.878	0.933	0.935	0.852	0.125
F†	42	0.673	0.886	0.841	0.928	0.878	0.841	0.099
	1337	0.648	0.887	0.879	0.913	0.910	0.847	0.112
G	42	0.666	0.876	0.876	0.919	0.895	0.846	0.103
	1337	0.646	0.882	0.875	0.916	0.915	0.847	0.114
H	42	0.663	0.880	0.832	0.915	0.915	0.841	0.105
	1337	0.672	0.883	0.842	0.913	0.912	0.844	0.101
I	42	0.677	0.885	0.873	0.912	0.910	0.851	0.099
	1337	0.704	0.877	0.824	0.915	0.888	0.841	0.084

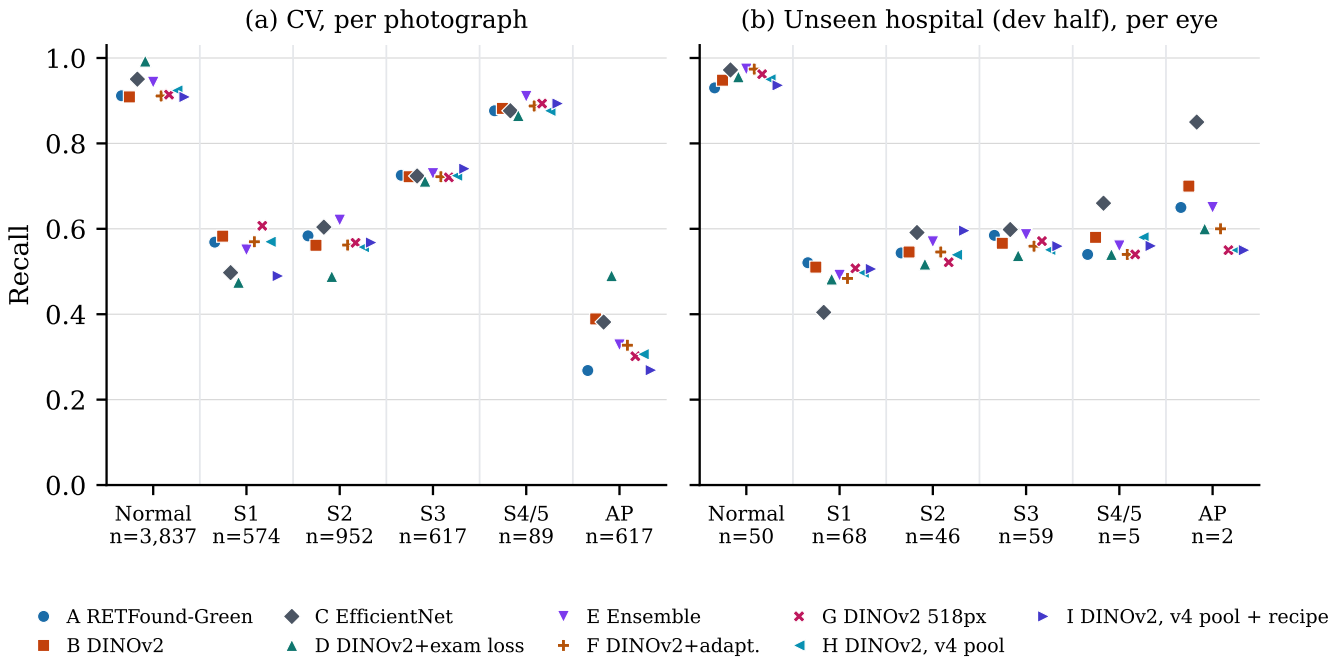


Fig. 4. Recall per class, all arms: (a) CV per photograph; (b) unseen hospital per eye. n below each class: photographs (a), eyes (b). Arm identity by colour and marker.

TABLE IV

LEAVE-ONE-SOURCE-OUT (LOSO), PER PHOTOGRAPH, SEEDS POOLED: QWK5, MF1 AND ACCURACY WITH 95% INFANT-CLUSTERED CI, AND MEAN RECALL PER CLASS (– ABSENT FROM THE HELD-OUT SOURCE). SHENZHEN IS THE PROTOCOL'S SITE FOLD; THE OSTRAVA AND ROP-VL FOLDS WERE ADDED BY AMENDMENT 5 FOR B, H AND I. POOL: THE TRAINING POOL (TABLE XV).

					Mean recall per class					
Arm	Pool	QWK5 [95% CI]	mF1 [95% CI]	Acc. [95% CI]	Normal	S1	S2	S3	S4/5	AP
Shenzhen held out (fold_site.csv): 728 photos, 396 infants, 12 duplicates excluded										
A	v3	0.958 [0.948, 0.966]	0.832 [0.799, 0.863]	0.867 [0.840, 0.892]	0.99	0.58	0.79	0.90	–	–
B	v3	0.962 [0.953, 0.971]	0.841 [0.809, 0.871]	0.882 [0.859, 0.906]	1.00	0.58	0.81	0.93	–	–
D	v3	0.958 [0.945, 0.968]	0.836 [0.804, 0.867]	0.877 [0.853, 0.900]	1.00	0.53	0.85	0.90	–	–
H	v4	0.960 [0.950, 0.969]	0.839 [0.808, 0.870]	0.877 [0.854, 0.901]	1.00	0.59	0.79	0.92	–	–
I	v4	0.942 [0.926, 0.956]	0.777 [0.741, 0.814]	0.840 [0.813, 0.868]	1.00	0.39	0.72	0.92	–	–
Ostrava held out (fold_loso_ostrava.csv): 4,285 photos, 148 infants, 0 duplicates excluded										
B	v3	0.740 [0.398, 0.845]	0.483 [0.384, 0.596]	0.791 [0.710, 0.868]	0.95	0.14	0.20	0.36	–	0.79
H	v4	0.766 [0.452, 0.858]	0.479 [0.380, 0.590]	0.797 [0.718, 0.876]	0.97	0.13	0.20	0.35	–	0.77
I	v4	0.779 [0.471, 0.875]	0.487 [0.394, 0.595]	0.804 [0.721, 0.880]	0.97	0.15	0.19	0.36	–	0.82
ROP-VL held out (fold_loso_ropvl.csv): 1,673 photos, 983 infants, 2 duplicates excluded										
B	v3	0.800 [0.771, 0.827]	0.452 [0.437, 0.467]	0.677 [0.653, 0.701]	0.94	0.74	0.60	0.71	0.00	0.00
H	v4	0.801 [0.774, 0.824]	0.443 [0.427, 0.458]	0.658 [0.634, 0.682]	0.89	0.71	0.60	0.74	0.00	0.00
I	v4	0.796 [0.768, 0.820]	0.462 [0.445, 0.479]	0.676 [0.652, 0.699]	0.94	0.48	0.67	0.81	0.00	0.06
Mean QWK5 over the three held-out sources (Amendment 5c): B 0.834, H 0.842, I 0.839										

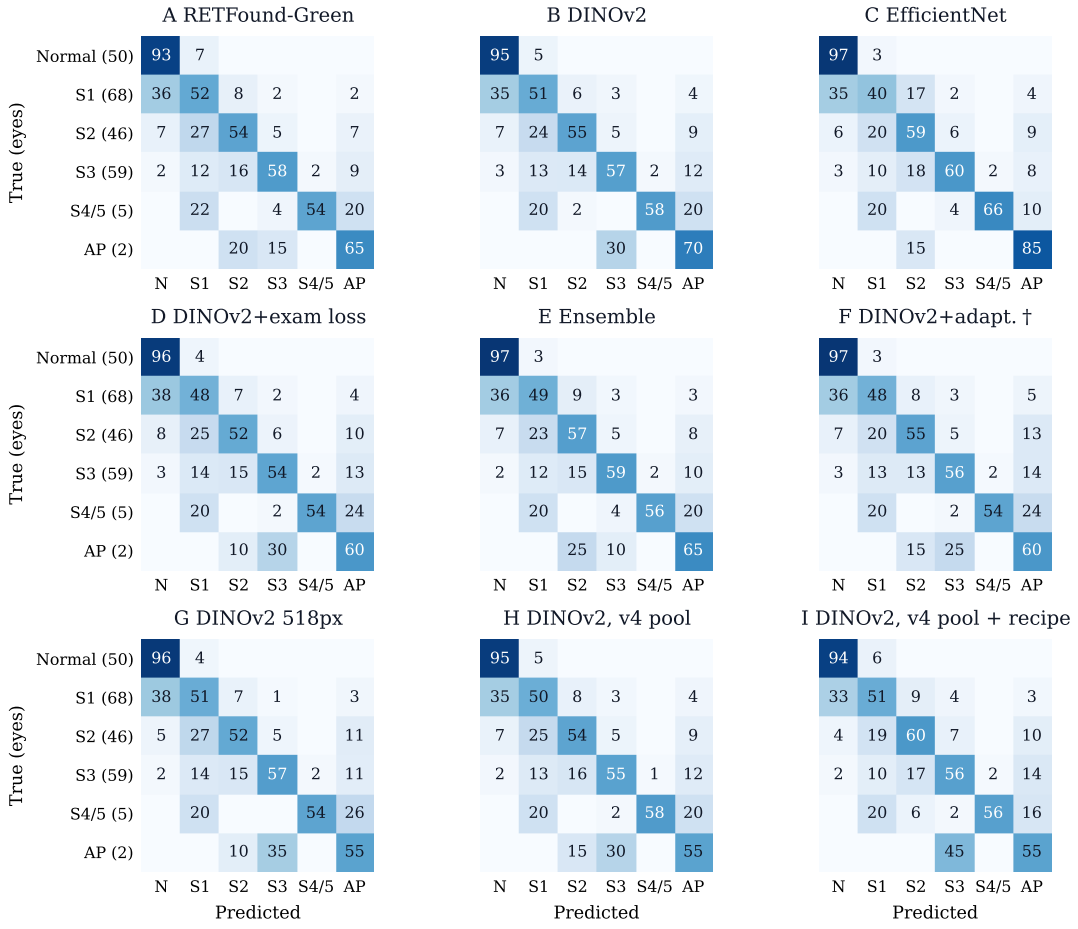


Fig. 5. Eye-level confusion matrices at the unseen hospital (dev half), all arms. Cell: share of the true-class row in %, averaged over the arm's 10 models (diagonal = recall); blank = 0. True-class eye counts in parentheses. N normal, S stage, AP AP-ROP. †Transductive.

TABLE V

ALL 12 PRE-SPECIFIED PAIRED CONTRASTS (FIRST ARM MINUS SECOND): QWK5 AND MF1, BOTH ARMS ON THE SAME INFANT RESAMPLE, 95% CI. CV: SELECTION CONTEXT; DEV: DESCRIPTIVE. NO ROW IS A CONFIRMATORY TEST; THE PRIMARY ENDPOINT (LOCKED-HALF EYE-LEVEL A−C) IS UNSCORED.

		CV, per photograph		Dev half, per eye	
Pair	Contrast	ΔQWK5 [95% CI]	ΔmF1 [95% CI]	ΔQWK5 [95% CI]	ΔmF1 [95% CI]
A−C	retinal vs CNN (primary)	−0.022 [−0.052, +0.008]	−0.013 [−0.025, +0.012]	+0.014 [−0.012, +0.048]	−0.009 [−0.063, +0.031]
A−B	retinal vs natural	−0.031 [−0.077, +0.015]	−0.010 [−0.023, +0.036]	+0.019 [+0.000, +0.041]	−0.003 [−0.032, +0.016]
B−C	natural vs CNN	+0.009 [−0.018, +0.027]	−0.003 [−0.041, +0.018]	−0.004 [−0.038, +0.030]	−0.007 [−0.066, +0.042]
E−C	ensemble vs CNN	+0.005 [−0.007, +0.019]	+0.008 [−0.001, +0.024]	+0.010 [−0.012, +0.040]	−0.005 [−0.058, +0.034]
E−A	ensemble vs retinal	+0.026 [+0.008, +0.046]	+0.021 [+0.007, +0.029]	−0.004 [−0.020, +0.008]	+0.004 [−0.010, +0.022]
D−A	exam loss vs retinal	+0.002 [−0.057, +0.046]	+0.018 [−0.074, +0.055]	−0.028 [−0.048, −0.011]	−0.022 [−0.047, +0.019]
D−B	exam loss vs per photo	−0.029 [−0.051, +0.003]	+0.008 [−0.042, +0.038]	−0.009 [−0.019, +0.001]	−0.024 [−0.042, −0.002]
F−B	adaptation	−0.014 [−0.035, +0.008]	−0.009 [−0.021, +0.021]	−0.012 [−0.022, −0.002]	−0.014 [−0.027, +0.006]
G−B	518 vs 392 px	−0.011 [−0.036, +0.012]	−0.006 [−0.020, +0.026]	+0.012 [−0.006, +0.032]	−0.014 [−0.037, +0.016]
H−B	v4 pool	−0.013 [−0.031, +0.009]	−0.007 [−0.017, +0.019]	−0.006 [−0.015, +0.003]	−0.010 [−0.023, +0.010]
I−B	v4 pool + recipe	−0.004 [−0.015, +0.007]	−0.035 [−0.053, −0.001]	+0.020 [+0.003, +0.041]	−0.005 [−0.057, +0.033]
I−H	v4 recipe	+0.009 [−0.007, +0.020]	−0.028 [−0.043, −0.010]	+0.026 [+0.006, +0.046]	+0.004 [−0.050, +0.045]

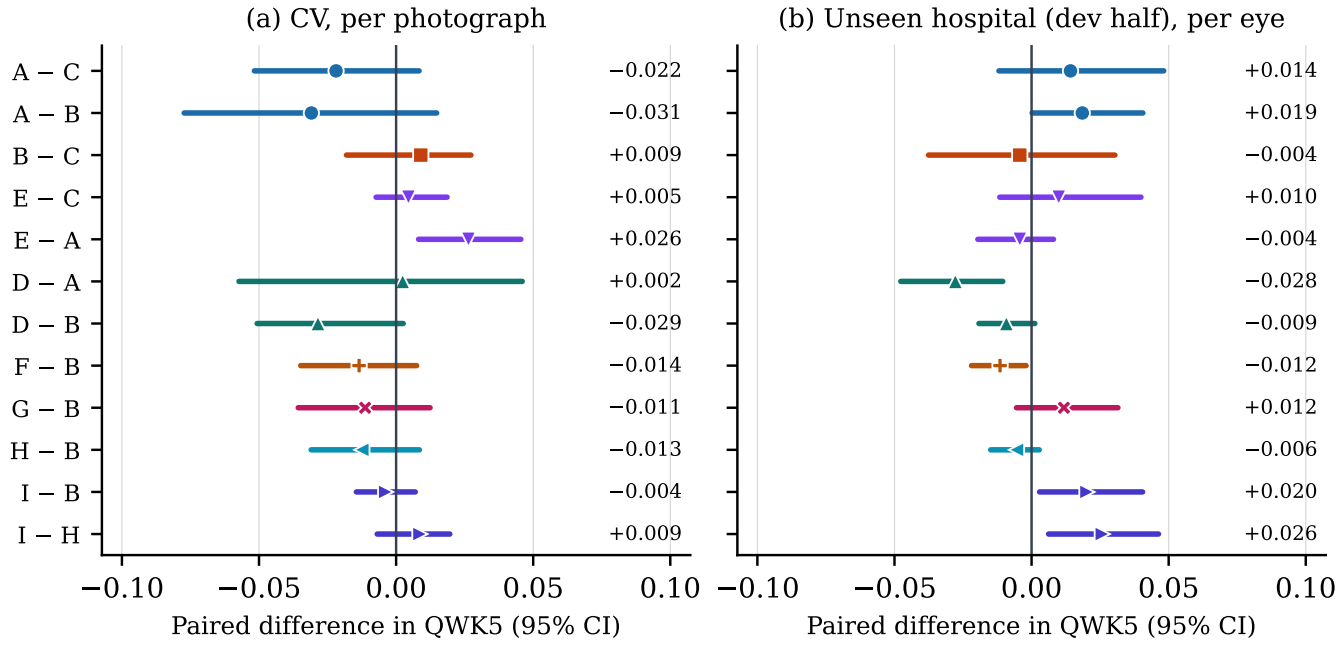


Fig. 6. Paired contrasts in QWK5 with 95% CI (Table V): (a) CV, 6,686 photographs; (b) dev half, 230 eyes. A line crossing zero is a tie; point differences at right.

TABLE VI

OPERATING POINT AND CALIBRATION, PER ARM AND SEED. THRESHOLD: THE LARGEST VALUE OF THE UNIT ANY-ROP SCORE, $\max_i(1 - p_{i,\text{normal}})$, THAT KEEPS CV UNIT-LEVEL ANY-ROP SENSITIVITY AT OR ABOVE 0.90; CARRIED UNCHANGED TO THE DEV HALF, WHERE SENSITIVITY AND SPECIFICITY ARE MEANS OVER THE SEED'S FIVE FOLD MODELS. SEVERE: STAGE 3 OR WORSE, INCLUDING AP-ROP, AT THE SAME THRESHOLD. T: TEMPERATURE FITTED ON CV OUT-OF-FOLD UNIT VECTORS. ECE: TEN BINS, DEV EYES, SEEDS POOLED, AS TRAINED AND AFTER T (FIG. 7).

			Any ROP vs normal				Severe (S3+) vs rest					Dev-eye ECE	
Arm	Seed	Threshold	CV sens	CV spec	Dev sens	Dev spec	CV sens	CV spec	Dev sens	Dev spec	T	as trained	with T
A	42	0.9884	0.900	0.999	0.650	1.000	0.952	0.652	0.924	0.659	0.840	0.152	0.205
	1337	0.9887	0.900	0.996	0.670	1.000	0.955	0.651	0.933	0.640	0.853		
B	42	0.9894	0.900	0.996	0.656	1.000	0.945	0.648	0.927	0.654	0.850	0.157	0.203
	1337	0.9893	0.900	0.998	0.676	1.000	0.957	0.654	0.942	0.638	0.851		
C	42	0.9850	0.900	0.991	0.680	1.000	0.948	0.646	0.894	0.613	0.917	0.183	0.208
	1337	0.9805	0.900	0.993	0.671	1.000	0.942	0.645	0.918	0.633	0.935		
D	42	0.9893	0.900	0.998	0.656	1.000	0.952	0.651	0.900	0.643	0.828	0.185	0.238
	1337	0.9876	0.900	0.999	0.639	1.000	0.948	0.650	0.909	0.665	0.838		
E	42	0.9866	0.900	1.000	0.650	1.000	0.945	0.650	0.912	0.654	0.747	0.135	0.214
	1337	0.9848	0.900	0.998	0.662	1.000	0.941	0.647	0.918	0.643	0.762		
F†	42	0.9891	0.900	0.996	0.679	1.000	0.944	0.647	0.921	0.626	0.853	0.184	0.223
	1337	0.9890	0.900	0.997	0.669	1.000	0.959	0.654	0.933	0.642	0.863		
G	42	0.9882	0.900	0.996	0.643	1.000	0.949	0.649	0.921	0.665	0.854	0.174	0.210
	1337	0.9897	0.900	0.996	0.667	1.000	0.955	0.651	0.939	0.646	0.868		
H	42	0.9884	0.900	0.996	0.658	1.000	0.946	0.648	0.918	0.648	0.836	0.167	0.215
	1337	0.9898	0.900	0.998	0.661	1.000	0.953	0.652	0.927	0.648	0.851		
I	42	0.9812	0.900	0.998	0.676	1.000	0.937	0.646	0.912	0.626	0.620	0.061	0.196
	1337	0.9816	0.900	0.998	0.680	1.000	0.938	0.646	0.918	0.623	0.631		

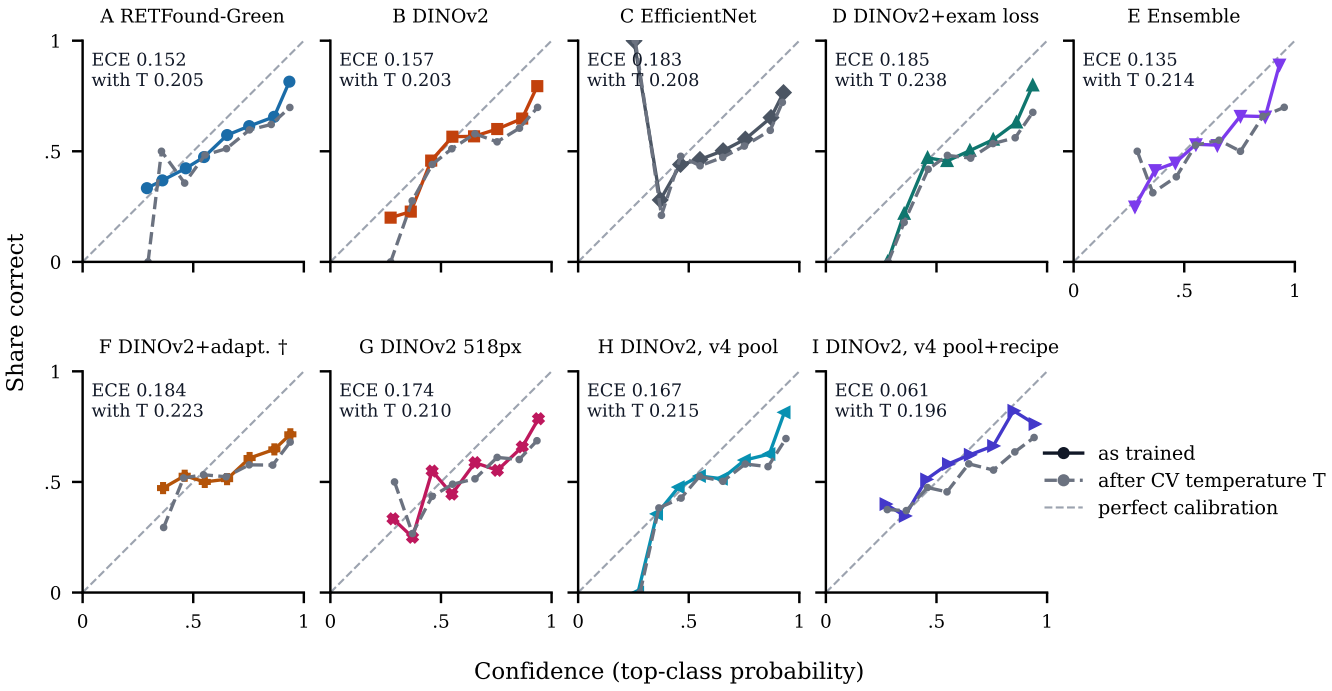


Fig. 7. Reliability of the eye-level top-class probability at the unseen hospital (dev half), ten equal-width bins, seeds pooled: as trained (colour) and after each seed's CV temperature T (grey). †Transductive.

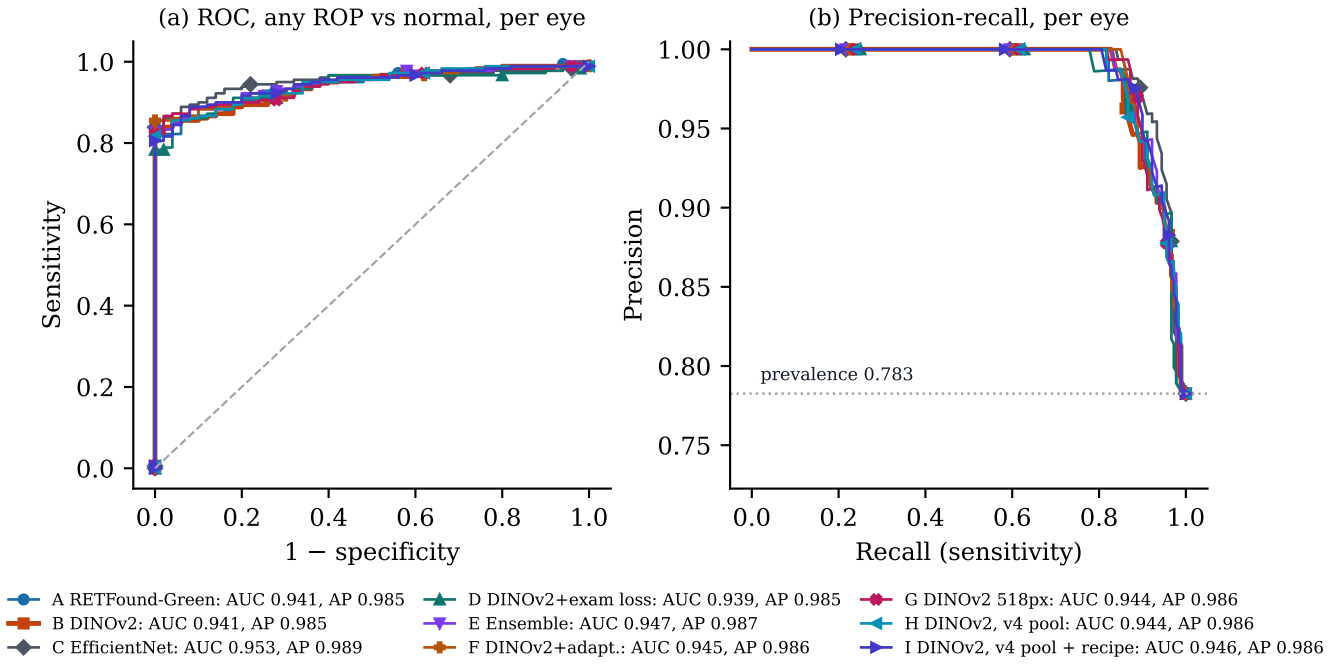


Fig. 8. Any ROP vs normal per eye at the unseen hospital (dev half), all arms. Curves use the eye score averaged over each arm's 10 models, so their AUC and AP (legend) differ slightly from Table I, which averages per-model values. Dotted: prevalence.

TABLE VII

RECIPE AND COST. SHARED BY EVERY TRAINED ARM: CIRCLE-CROP, LETTERBOX, CLAHE; CLASS-BALANCED CROSS-ENTROPY [6]; STRONG AND SITE-RANDOMISING AUGMENTATION; BATCH 16; ADAMW; ONE-CYCLE COSINE SCHEDULE; LAST-EPOCH WEIGHTS. LLRD: LAYER-WISE LR DECAY; LS: LABEL SMOOTHING. F'S ADAPTATION IS SELF-ENSEMBLING [7]. COST: MEAN OVER CV FOLDS 0–4 AND BOTH SEEDS ON KAGGLE TESLA T4 GPUS (KERNEL LOGS); C AND E WERE NOT TRAINED IN THIS GENERATION.

Arm	Backbone	px	Norm.	Pool	Peak lr	LLRD	LS	Ep.	Differences from B's recipe	min/fold	img/s
A	`retfound_green`	392	half	v3	2e-4	0.8	0.05	20	RETFound-Green weights	32.1	38.6
B	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	reference recipe	33.3	31.9
C	`effnetv2s`	384	imagenet	v3	3e-4	–	0.05	20	v2 flat softmax, natural sampler; saved predictions re-scored	–	–
D	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	LSE bag loss on Ostrava sessions, K=8, τ=1	32.5	29.0
E	–	–	–	v3	–	–	–	–	mean of A, B, C probabilities, matched by seed; no training	–	–
F†	`dinov2_s`	392	imagenet	v3	1e-4	0.8	0.05	20	self-ensembling on unlabelled dev photos, EMA 0.99, conf. $\geq$0.9	61.7	14.2
G	`dinov2_s`	518	imagenet	v3	1e-4	0.8	0.05	20	518 px	58.1	19.2
H	`dinov2_s`	392	imagenet	v4	1e-4	0.8	0.05	20	v4 pool (per-device quality filter)	34.1	31.2
I	`dinov2_s`	392	imagenet	v4	1e-4	0.8	0.1	20	EMA weights 0.999; 1 head-only epoch; label smoothing 0.1; rot. $\pm$180°, crop $\geq$0.85, gamma, noise, Fourier swap p 0.5	34.8	32.1

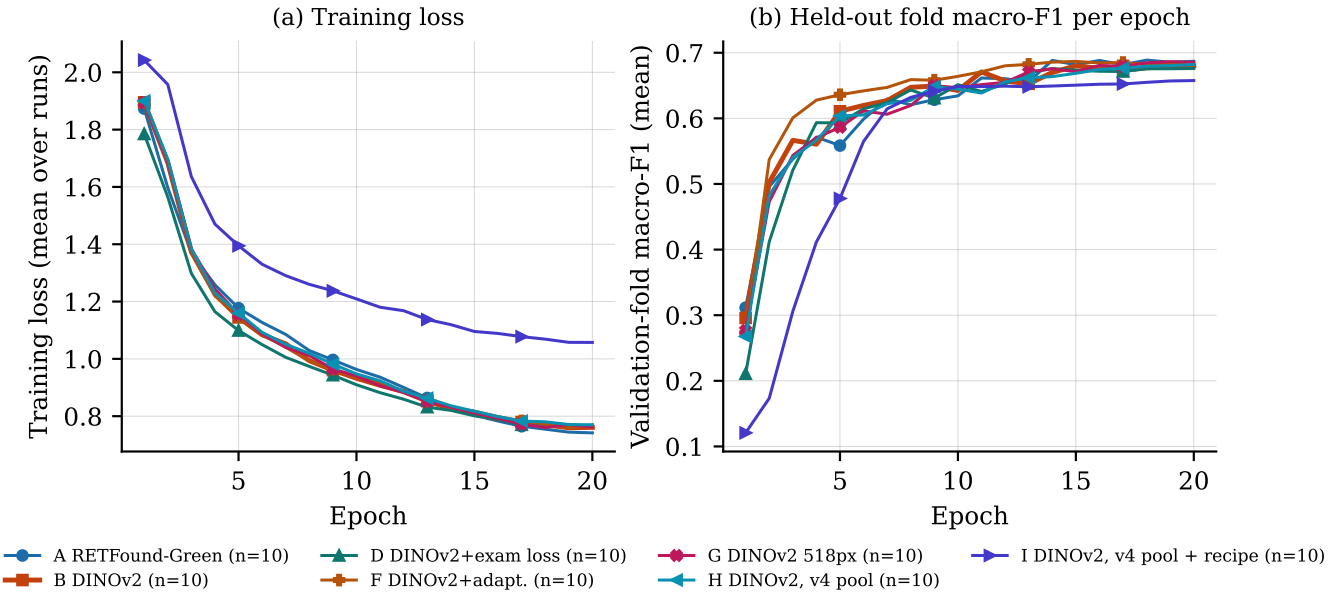


Fig. 9. Training curves, mean over each arm's 10 runs (five CV folds × two seeds): (a) training loss; (b) macro-F1 on the held-out fold per epoch. I's loss sits higher by construction (its label smoothing, Table VII) and its F1 is measured on EMA weights. C and E have no curves.

TABLE VIII

SECONDARY READOUTS. DEV PER PHOTOGRAPH: QWK5 WITH 95% CI, NO EYE AGGREGATION. CV PER COHORT AT ITS LABEL UNIT: OSTRAVA EXAMINATION SESSION, ROP-VL EYE EXAMINATION, SHENZHEN PHOTOGRAPH [5]. ORDINAL DECODER (AMENDMENT 3B): EXPECTED STAGE CUT BY FOUR THRESHOLDS FITTED ON CV ONLY; ARGMAX REMAINS PRIMARY.

	Dev half	CV, each cohort at its label unit			Ordinal decoder	
Arm	per photo QWK5 [95% CI]	Ostrava exam	ROP-VL eye-exam	Shenzhen photo	CV	Dev eye
A	0.407 [0.336, 0.478]	0.679	0.931	0.969	0.757	0.761
B	0.416 [0.342, 0.488]	0.701	0.930	0.967	0.788	0.760
C	0.438 [0.362, 0.510]	0.775	0.921	0.967	0.815	0.771
D	0.412 [0.339, 0.484]	0.846	0.925	0.967	0.808	0.762
E	0.418 [0.346, 0.492]	0.769	0.940	0.971	0.812	0.776
F†	0.416 [0.342, 0.490]	0.693	0.935	0.967	0.783	0.748
G	0.413 [0.343, 0.487]	0.695	0.925	0.967	0.779	0.764
H	0.412 [0.338, 0.486]	0.710	0.933	0.968	0.789	0.762
I	0.441 [0.363, 0.519]	0.712	0.933	0.962	0.807	0.777

TABLE IX

v1 (JUNE 2026): EFFICIENTNET-B3, ONE SHARED MODEL FOR BINARY ROP AND DR. P / R: PRECISION / RECALL. *CAVEAT*: IMAGE-LEVEL SPLIT AND TWO DISEASES IN ONE MODEL; RETIRED.

Split	n	Acc.	mF1	No ROP P / R	ROP P / R	ROP F1
ROP val	928	0.647	0.646	0.476 / 0.933	0.941 / 0.510	0.661
ROP test	1,502	0.733	0.731	0.644 / 0.913	0.891 / 0.585	0.706
DR val, ep. 5	–	0.719	0.497	–	–	–

TABLE X

JUNE 2026 BINARY SWEEP (ROP YES/NO, OSTRAVA, SWEEP VALIDATION SPLIT; QWK_2 BINARY KAPPA), AND, *, A SEPARATE RESNET50 RUN ON ITS OWN VALIDATION AND TEST SPLIT (`RESULTS/ROP/METRICS.JSON`). *CAVEAT*: INFLATED BY A DEVICE SHORTCUT (IMAGE DIMENSIONS ALONE PREDICTED ROP WITH AUC 0.911).

Architecture	Acc.	mF1	QWK_2	AUC	Params (M)	CPU ms
efficientnet_b0	0.758	0.757	0.525	0.867	4.01	56.2
efficientnet_v2_s	0.706	0.701	0.431	0.870	20.18	130.1
mobilenet_v3_large	0.744	0.743	0.500	0.857	4.2	34.4
resnet50	0.886	0.882	0.766	0.964	23.51	157.2
densenet121	0.764	0.764	0.537	0.887	6.96	136
resnet50* val	0.747	0.740	0.499	0.842	–	–
resnet50* test	0.844	0.843	0.688	0.927	–	–

III. EARLIER GENERATIONS (HISTORICAL)

Each table keeps its own protocol and definitions; none is comparable with Tables I–VIII without its caveat. Glaucoma and diabetic-retinopathy models (`MODEL_HISTORY.md`) are outside this ROP compendium.

IV. DATA

Ostrava [8] labels one examination session; ROP-VL one photograph with an eye-examination identifier; Shenzhen one photograph; Multi-View one eye.

V. REPRODUCIBILITY

Figures, tables and every number above: `.venv/bin/python scripts/paper3_figures.py`, then `tectonic -X compile paper3_results.tex`. No model is run and no file named `test_locked` is read. PKG = `FINAL_MODEL_EVIDENCE_PACKAGE`.

COMPLIANCE WITH ETHICAL STANDARDS

This retrospective study used only publicly released, de-identified datasets under their licences (Ostrava/RIDIRP, ROP-VL, Shenzhen, Multi-View ROP, CC BY-NC); no new data were collected and no ethical approval was required. RETFound-Green's weights are used under their non-commercial licence. The system is research software and not a medical device. No funding was received; the authors declare no conflicts of interest.

ACKNOWLEDGMENTS

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TABLE XI

v2 (August 2026): EfficientNetV2-S structured head (CORN ordinal + AP-ROP branch + site adversary, 20,512,600 parameters) vs flat softmax, natural sampling, five-fold CV. QWK_6: legacy six-class definition; $QWK5_d$: v2's "corrected" definition, rows with true or predicted AP-ROP removed (optimistic); QWK5: the v3 definition used in Table I. Dev and locked: unseen hospital; eye = max over views; locked half opened once (2026-08-18); dev and locked values are means over the five fold checkpoints. The served fold-0 checkpoint (structured, seed 42) scored per photograph $QWK5_d$ 0.493, accuracy 0.546, any-ROP AUC 0.835, and per eye $QWK5_d$ 0.839 [0.768, 0.890], accuracy 0.678, mF1 0.616. *Caveat*: arm C re-scores the flat arm under v3 rules; v2's headline eye number was on the locked half, v3's on the dev half.

		CV pooled OOF, 6,700 photos						Dev	Locked, photo		Locked, eye (max)		
Head	Seed	QWK_6	$QWK5_d$	QWK5	Acc.	mF1	AUC	QWK5	$QWK5_d$	QWK5	$QWK5_d$	QWK5	Acc.
Structured	42	0.761	0.897	0.807	0.786	0.652	0.969	0.755	0.473	0.517	0.813	0.823	0.646
	1337	0.783	0.905	0.824	0.804	0.666	0.967	0.752	0.456	0.500	0.794	0.808	0.627
Flat	42	0.777	0.897	0.820	0.789	0.655	0.968	0.756	0.492	0.551	0.808	0.823	0.635
	1337	0.770	0.898	0.813	0.787	0.658	0.966	0.764	0.486	0.543	0.801	0.818	0.630

TABLE XII

v2 five-backbone staging sweep (flat softmax, six classes, fold 0 and the leave-site fold, seed 42). *Caveat*: one fold and one seed; the in-fold spread lies inside one run's end-of-schedule noise; legacy QWK not recomputed.

Backbone	Params	mF1, fold 0	mF1, site fold	AP-ROP recall
EfficientNetV2-S	20.2M	0.520	0.641	0.183
DeiT3-S	21.8M	0.517	0.507	0.174
ConvNeXt-T (in12k)	27.8M	0.509	0.530	0.162
CAFormer-S18	24.3M	0.498	0.507	0.221
EfficientNet-B0	4.0M	0.492	0.536	(pre-metric-fix)

TABLE XIII

Retired v2 structured arm trained with a class-balanced sampler, fold 0. *Caveat*: fold 0 only, legacy six-class QWK; the CORN cascade calibrates to the training prevalence.

Fold 0	Balanced sampler	Natural (v2)	Flat
QWK (legacy 6-class, as reported at the time)	0.121	0.569	0.594
stage_4_5 precision	0.043	—	—
Normal recall	0.456	0.939	—
macro-F1	0.403	0.553	0.563

TABLE XIV

Retired ResNet50 binary screener: external audit at the unseen hospital (663 dev photographs). *Caveat*: its threshold was set at another hospital; the v2 dev AUC is in-sample for v2's own threshold choice.

External audit	Value
images flagged positive	663 of 663 — including every healthy eye
specificity at the served threshold	0.000
external AUC	0.691
minimum score among external healthy eyes	0.2939
served threshold	0.1933
external dev AUC ("any ROP") (retired / v2 structured)	0.691 / 0.821
external locked AUC ("any ROP") (retired / v2 structured)	not measured / 0.835

TABLE XV

DATA BY HOSPITAL AND CLASS. CV POOL: EVERY PHOTOGRAPH LIES IN EXACTLY ONE VALIDATION FOLD. TRAINING-ELIGIBLE: AFTER PER-INFANT CAPS (TRAINING SIDES ONLY); THE V4 POOL ADDS A PER-DEVICE OSTRAVA QUALITY FILTER. *LOCKED-HALF LABELS WERE NOT READ; COUNTS FROM THE PROTOCOL.

Set	Unit	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP	Total	Infants
Cross-validation pool (every photo is in exactly one validation fold)									
Ostrava	photos	2,980	252	458	125	0	470	4,285	148
ROP-VL	photos	623	236	338	241	89	148	1,675	983
Shenzhen	photos	234	89	163	254	0	0	740	397
Total	photos	3,837	577	959	620	89	618	6,700	1,528
Training-eligible after per-infant caps, v3 pool (arms A–G)									
Ostrava	photos	270	190	195	40	0	20	715	
ROP-VL	photos	623	236	338	241	89	148	1,675	
Shenzhen	photos	216	89	163	254	0	0	722	
Total	photos	1,109	515	696	535	89	168	3,112	
Training-eligible, v4 pool (arms H, I)									
Ostrava	photos	270	173	191	40	0	20	694	
ROP-VL	photos	623	236	338	241	89	148	1,675	
Shenzhen	photos	216	88	158	252	0	0	714	
Total	photos	1,109	497	687	533	89	168	3,083	
Unseen hospital (Multi-View), never in any fold									
Dev half	photos	150	190	135	168	14	6	663	116
Dev half	eyes	50	68	46	59	5	2	230	116
Locked half*	photos	–	–	–	–	–	–	661	116
Locked half*	eyes	–	–	–	–	–	–	230	116

TABLE XVI

LOSO FOLD COMPOSITIONS: HELD-OUT SOURCE (TEST PHOTOGRAPHS AFTER DUPLICATE EXCLUSION, INFANTS) AND TRAINING PHOTOGRAPHS BY CLASS IN THE V3 POOL (A, B, D) AND THE V4 POOL (H, I).

					Training photographs						
Held out	Fold file	Test photos	Infants	Pool	Normal	S1	S2	S3	S4/5	AP	Total
Shenzhen	fold_site.csv	728	396	v3	893	426	533	280	89	167	2,388
				v4	893	409	529	280	89	167	2,367
Ostrava	fold_loso_ostrava.csv	4,285	148	v3	839	322	494	492	89	147	2,383
				v4	839	322	494	492	89	147	2,383
ROP-VL	fold_loso_ropvl.csv	1,673	983	v3	486	276	351	292	0	20	1,425
				v4	486	259	347	292	0	20	1,404

TABLE XVII

SOURCES: PATH AND LAST COMMIT TOUCHING IT (`GIT LOG`); "GIT-IGNORED, ON DISK" FILES EXIST ONLY IN THE WORKING TREE (THE CSVS FALL UNDER THE REPOSITORY'S `*.CSV` RULE).

Item	Path	Commit
Protocol and amendments (first commit)	`docs/paper/PROTOCOL_V2.md`	`ac33f29` 2026-10-20
Protocol, latest amendment	`docs/paper/PROTOCOL_V2.md`	`0e55915` 2026-10-31
Scores, arms A–I (selection, LOSO, contrasts)	`results/rop/v3/v3_scores.json`	`b827cce` 2026-11-01
Per-arm report data (tables, curves)	`results/rop/v3/report_data.json`	`b827cce` 2026-11-01
v4 pool summary	`results/prepared/rop_staging/v4/v4_summary.json`	`1efccd0` 2026-10-31
Scorer	`scripts/rop_v3_score.py`	`2bbc55a` 2026-10-31
Report-data builder	`scripts/rop_v3_report_data.py`	`d8b7ec1` 2026-10-31
v4 model report	`scripts/rop_v4_model_report.py`	`3f7b79f` 2026-11-01
This paper's figures and tables	`scripts/paper3_figures.py`	not yet committed
Model chronology	`docs/paper/MODEL_HISTORY.md`	`11b8f5e` 2026-10-31
v2 QWK recalculation (3 definitions)	`PKG/20_AUDIT/verification_pass_2/06_VALIDATION/qwk_5class_vs_6class_recalculation.csv`	git-ignored, on disk
v2 CV summary	`PKG/07_TEST_RESULTS/metrics_summary.csv`	git-ignored, on disk
v2 locked, served checkpoint (eye CIs)	`PKG/08_EYE_LEVEL_ANALYSIS/eye_confidence_intervals.csv`	git-ignored, on disk
Five-backbone sweep	`PKG/19_COMPARISONS/retired_models/five_backbone_benchmark.md`	`aaffcf0` 2026-09-22
Weighted-sampler arm	`PKG/19_COMPARISONS/retired_models/weighted_sampler_v1_arm.md`	`aaffcf0` 2026-09-22
ResNet50 screener audit	`PKG/19_COMPARISONS/retired_models/resnet50_binary_screener.md`	`aaffcf0` 2026-09-22
June 2026 binary sweep	`results/sweep_rop.md`	`2bb8db3` 2026-06-28
ResNet50 val/test metrics	`results/rop/metrics.json`	`c042d7f` 2026-07-31
v1 EfficientNet-B3 evaluation	`junk/unified_model/evaluation_unified.txt`	git-ignored, on disk
Arm configs (A, B, D, F–I; C)	`configs/rop_staging_*.yaml`	`9c9ab65` 2026-10-31